

Scalability & Future Plan

Current limitations, scalability strategy and future roadmap.

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Scalability & Future Plan

The current solution is a Node.js/Express REST API with MongoDB persistence, JWT authentication and Gemini-powered AI features. The following plan identifies practical future improvements without claiming that they are already implemented.

Current System Limitations

ID	Limitation	Impact	Priority
1	No deployed application or production URL supplied	Production-scale behavior cannot be verified from the supplied project materials.	High
2	Performance measurements are not yet supplied	Actual throughput and response-time capacity remain unverified.	Medium
3	Full source-code folder and database seed files were not supplied	Source-level verification and repeatable data setup are limited.	Medium

Scalability Plan

ID	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Express API with JWT-protected routes.	Introduce load balancing, horizontal scaling and rate limiting as traffic grows.
2	Data Storage	MongoDB/Mongoose persistence.	Use indexing, query optimization, connection tuning and, if required, database scaling/replication.
3	Performance	API and AI endpoints are defined; measured performance is pending.	Add caching where appropriate, background processing for long AI jobs, monitoring and load testing.
4	Security	JWT authentication, bcrypt and environment-based secrets are documented.	Add stronger production secret management, rate limits, security headers, audit logging and regular dependency updates.

Future Roadmap

Phase	Planned Features / Enhancements	Target Timeline	Expected Impact
Phase 1	Complete source-level verification, testing and deployment preparation.	Next milestone	Improved submission readiness and reproducibility.
Phase 2	Performance monitoring, load testing and API optimization.	Future sprint	Better visibility into response time and throughput.
Phase 3	Scalable deployment, caching and database optimization.	Future release	Support for higher traffic and data volumes.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 4	Extended AI features, observability and production hardening.	Long-term	Broader functionality and stronger operational reliability.